Accelerated Quantum Supercomputing in Action: A Hands-On Tutorial on Scalable Hybrid Workflows

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I. Detailed Outline

Date: Sun Nov 16, 1:30 - 5pm

- **Hour 1 : Accelerated Quantum Supercomputing (NV, NERSC)**
  - ▷ Onboarding with NERSC
    - ○ Logging in, running interactive sessions, submitting jobs, and post-processing
    - ○ Using Jupyter notebooks on Perlmutter using a previously tested container
  - ▷ Introduction to AQSC
    - ○ Overview
    - ○ Emphasis on hybrid workflows
    - ○ Mention of QEC efforts with Infleqtion
  - ▷ Hands-on with CUDA-Q
    - ○ Introductory Jupyter notebooks with simple hybrid workflows such as VQE
    - ○ Different backends for executing quantum circuits
    - ○ CUDA-QX intro
    - ○ Scaling simulations from 1 to 100s of GPUs across nodes with minimal changes in workflow.
- **Hour 2: AI for Scalable Quantum Algorithms (NV, NERSC)**
  - ▷ Hybrid algorithms
    - ○ Introduce GQE
    - ○ Compare and contrast with VQE
  - ▷ GQE hands-on
    - ○ Algorithm design
    - ○ Parallelization opportunities
  - ▷ Advanced
    - ○ Quickly highlight AFQMC, QAOA-GPT, etc.
    - ○ Additional resources
- **Hour 3: Logical Qubit Experiments (Infleqtion, NERSC)**
  - ▷ 1. Running a logical qubit VQE ground state prep with `cudaq`:
    - ○ Materials science VQE application (single-impurity Anderson model) in CUDA-Q using two logical qubits. The demo is contained in a Jupyter notebook for generating and noisily simulating logically encoded circuits with the [[4, 2, 2]] code. Users can use a GPU to leverage the CUDA-Q GPU backend for the Hamiltonian Variational Ansatz optimization in `cudaq-solvers`.
  - ▷ 2. Running decoding for the many-hypercube QEC code using `cudaq-qec` and Infleqtion's `qLDPC` with NERSC's Perlmutter Supercomputer:

- Users will be able to execute a Python script to run decoding of real data from Infleqtion's Sqale QPU. In particular, the script will decode the [[16, 4, 4]] many-hypercube logical state preparation reported in `https://arxiv.org/abs/2509.13247`. It uses `cudaq-qec`'s GPU-accelerated belief propagation (BP) decoder, along with the `qLDPC` library, to demonstrate a pipeline for decoding syndromes and retrieving logical state outcomes. We will show how to set up the environment for running the decoder on a GPU on Perlmutter in interactive mode.

▷ 3. Running an atom rearrangement experiment on Infleqtion's Sqale QPU:

- A Jupyter notebook will showcase the atom rearrangement capability of Infleqtion's Sqale QPU, a feature leveraged in executing Infleqtion's logical qubit experiments. Users will be able to submit a 2D bitmap array job encoding a specified pattern via the presence or lack of an atom in the Sqale QPU array. To do so, users will be able to acquire a Superstaq API key to gain cloud access to the QPU. Instructions to do so can be found here. Atom array capture results will be sent to the email associated with the user's API key, with lead time subject to the QPU's queue.